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Inventor(s)

Monetta; Anthony Louis et al.

BOAT STORAGE SYSTEM FOR BOATING ACCESSORIES

Abstract

A boat storage system for boating accessories, the boat storage system including a containment assembly having a row of upwardly opening pocket or pockets for receiving boating accessories, a plurality of straps are positioned on the containment assembly at a back side and/or lateral sides, the containment assembly removably attached to a framework of a ski tow bar facing toward the aft of the boat and elevated off of the surface from which the ski tow bar extends. The containment assembly can be filled with towels, boat fenders, life jackets, skiing accessories. The containment assembly having a plurality of pockets with one or more of the pockets having an expanded and a collapsed state and being retained in the collapsed state providing an adjustable volume storage and transport device.

Inventors: Monetta; Anthony Louis (Mishawaka, IN), Kniller; Andrew M. (Elkhart, IN), Borchardt; Aaron (Andover, MN)

Applicant: Polaris Industries Inc. (Medina, MN)

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Background/Summary

PRIORITY APPLICATIONS [0001] This application claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 63/559,578, filed Feb. 29, 2024, the content of which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] Current stowage solutions on boats for accessories such as boat fenders, life jackets, tow ropes, are typically cabinets with doors, or compartments with hatches in the deck, all typically with latches. Improvements in stowage solutions for such accessories would be welcomed by boating enthusiasts.

SUMMARY OF THE INVENTION

[0003] The inventors have recognized that current storage solutions for pontoon boats, or single hull boats associated with boating accessories, particularly boating accessories such as boating fenders, life jackets, are not particularly convenient for accessing, locating, and stowing such accessories, and do not provide any means of facilitating easy transport on and off the boat. Conventional stowage means of such boating accessories is in cabinets, under seats, and/or under deck compartments having hatches. All of these stowage solutions have latches and appropriately provide no visibility to contents when the cabinet door or hatch is closed, and often provide only limited visibility when the cabinet door or hatch is opened. In many cases, the inconvenience of stowing such boating accessories on the boat in the stowage compartments results in such accessories not being stowed at all. This can result in leaving the accessories exposed to the elements and if at publicly accessible docks, subject to removal by others. Also, pulling accessories out of compartments, when the boat is taken out, if they are not put to immediate use, means the accessories can clutter the passenger region which can not be desirable. The inventors have recognized such conventional stowage in cabinets and compartments in boats is often scattered about the boat, and particularly when the boat is not used regularly, or is used by different people, knowing what is on board as to these boating accessories, where they are located, and accessing them can be challenging.

[0004] Many pontoon and single hull recreational boats, have ski tow bars. The ski tow bars often having an upper generally horizontal tubular crossing portion extending in a port to starboard direction, with a ski rope attachment fixture attached or attachable at a midpoint of the crossing portion of the ski tow bar. A pair of tubular portions extending downwardly from the crossing portion to the mounting flanges are typically attached to the deck of the boat. The ski tow bar can be positioned at the rearward periphery of pontoon boat decks attached to the top of the pontoon deck. This is normally at the fringe of or outside of the normal passenger region of pontoon boats. Similarly, in single hull boats the ski tow bar is normally in a rearward position with respect to the passenger area and can be attached to the deck or other boat structure elevated above the deck.

[0005] In embodiments of the invention, a tow bar containment assembly is expandable-collapsible and has a plurality of flexible sheet material wall portions extending from a base portion. The plurality of wall portions defines a plurality of large volume pockets, which can be four-sided pockets, where each pocket has upward pocket openings. The containment assembly can have a plurality of adjustable length straps positioned at the top margin, the side margins, or displaced from the margins on a back side of the containment assembly, where the plurality of adjustable length straps are attachable to the tow bar, and where the stowage containment assembly can be mounted on the forward side of the tow bar.

[0006] A feature and advantage of embodiments of the containment assembly is that it can be

positioned off the deck when desired such that walking deck space is minimally infringed upon. Elevating the storage allows the deck to dry or be readily cleaned, or allows other boating accessories, containers, or wearing apparel to be placed below the containment assembly.

[0007] A feature and advantage of the storage disclosed herein is that it makes access to accessories more convenient. A further feature and advantage of embodiments is that removal of boating accessories, such as boating fenders and life jackets, from the top opening pockets of the containment assembly is much easier than pulling accessories out of cabinets or other closed storage areas on boats. Boaters do not need to bend over to access the boating accessories from the containment assembly when the containment assembly is attached to the ski tow bar and elevated off of the deck.

[0008] A feature and advantage of embodiments is providing a convenient means for accessing, locating, and storing boating accessories such as boat fenders, towels, and life jackets on a boat, and providing a convenient and efficient means of transporting such accessories on and off the boat, minimizing such trips, all provided by a high-capacity containment assembly that is attachable at such convenient places as a ski tow bar or pontoon boat fencing. In embodiments, the containment assembly is reduceable in volume allowing different volumetric capacities of loads, while keeping the carried items generally snug within the containment assembly providing more secure containment of the contents being stored or transported.

[0009] A feature and advantage of embodiments is that stored boating accessories such as boating fenders, life jackets, and towels, are easily visible from passenger regions of the boat when the containment assembly filled with the boating accessories is attached to the ski tow bar and is elevated off of the deck.

[0010] A feature and advantage of embodiments is that the storage solution (e.g., the containment assembly) is soft sided, providing deformable soft contact surfaces when passengers bump or rub against the containment assembly.

[0011] In embodiments, a boat storage system has a ski tow bar with a ski tow bar framework, the framework having a generally horizontal upper tubular portion supported by a plurality of upright tubular support portions, and further has a containment assembly having a containment portion and an attachment portion integrated with the containment portion. The containment portion has one or more pockets formed of a flexible material and the attachment portion has at least one adjustable strap extending about and removably secured to ski tow bar framework.

[0012] In embodiments, a boat storage system is a boat with a metal framework having a generally horizontal upper framework portion supported by a plurality of upright framework support portions, and a soft sided containment assembly mounted to the generally horizontal upper tubular member. The containment assembly has a containment portion formed of a flexible material and the containment portion has an upward opening for receiving boating accessories in the containment portion. The soft sided containment assembly further has a flexible attachment portion attached to an upper edge portion of the containment portion. The flexible attachment portion is secured to and extends over the generally horizontal upper framework portion. The flexible attachment portion is securable to the framework and removable from the framework manually without tools.

[0013] In embodiments, a boat storage system is for stowing boating accessories. The boat storage system is a boat with a metal framework extending upwardly above a deck of the boat. The framework has an upper framework portion elevated above the deck, where the upper framework portion is supported by a plurality of upright support framework portions, and further has a soft sided containment assembly having a containment portion with a plurality of pockets. The containment assembly has an attachment portion attached to the containment portion, where the attachment portion has one or more adjustable length buckleable upper straps and a plurality of adjustable length buckleable lateral straps. The one or more adjustable length buckleable upper straps are secured to the upper framework portion and one or more of the plurality of adjustable

length buckleable lateral straps are secured to one or more of the plurality of upright support framework portions.

[0014] In embodiments, a boat storage system with boating accessories stowed therein, is a boat with a metal framework extending upwardly above a deck of the boat, the framework having an upper framework portion elevated above the deck, the upper framework portion supported by a plurality of upright support framework portions. A soft sided containment assembly has a containment portion and an attachment portion. The containment portion holds the boating accessories and the attachment portion secures the containment assembly to the metal framework.

[0015] In embodiments, a boat storage system for boating accessories has a containment assembly formed of flexible sewn fabric. The containment assembly has a row of three pockets including two lateral pockets and a middle pocket, each of the pockets having an upper opening. The containment assembly has an expanded state with each of the three pockets expanded to a full capacity for receiving the boating accessories, and has a collapsed state with at least the two lateral pockets in a folded-in state, the two lateral pockets retainable in the folded-in state. The row pockets of the containment assembly, when in the expanded state, have a width of at least 22 inches, a height of at least 13 inches, and a depth of at least 8 inches. The containment assembly further has a plurality of upper attachment straps proximate an upper margin of a back side of the sewn containment assembly for attachment of the containment assembly to a horizontal portion of a ski tow bar.

[0016] In embodiments, a boat storage system for boating accessories has a containment assembly attached to a ski tow bar on a boat, the containment assembly has a row of a plurality of compartments with open tops for receiving boating accessories, a plurality of upper attachment straps extending from a back side of the containment assembly to the ski tow bar, and a plurality of lower attachment straps extending from the back side of the containment assembly below the upper attachment straps to the ski tow bar. The containment portion has a volumetric capacity defined by the plurality of compartments of at least about 2 cubic feet. The containment assembly is elevated at least about 5 inches above boat structure to which the ski tow bar is attached.

[0017] In embodiments, a boat storage system for boating accessories is a boat with a tow bar and an expandable-collapsible boating accessory containment assembly removably attached to the tow bar. The tow bar has an upper horizontal tubular portion extending in a port-starboard direction, a pair of downward tubular portions extending to and being attached to the boat, a ski rope attachment fixture attached to or attachable to a center location on the upper horizontal tubular portion. The expandable-collapsible containment assembly has a fully expanded state and one or more collapsed states. The containment assembly includes a plurality of exterior wall portions and one or more interior divider wall portions, each of the exterior wall portions attached to and extending from a bottom side wall portion, and wherein each of the plurality of exterior wall portions and interior divider wall portions are formed of a flexible fabric. The plurality of exterior wall portions and interior divider wall portions define a plurality of five-sided pockets, each five-sided pockets having a respective upward pocket opening, wherein when the containment assembly is in the fully expanded states the volume of each of the plurality of the five sided pockets having a volume of at least about 0.4 cubic foot, and one of the exterior wall portions defines an attachment side wall portion of the containment assembly. The containment assembly also has a plurality of upper attachment straps extending from an upper region of the attachment side wall portion. The plurality of upper attachment straps are removably attached to the upper tubular horizontal portion. The containment assembly also includes a plurality of lower attachment straps extending from one or more of the plurality of exterior wall portions, where the plurality of lower attachment straps are each attached to the tow bar below the upper horizontal tubular portion.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0018] FIG. **1** is a perspective view of a pontoon boat.

[0019] FIG. **2** is a top plan view of a pontoon boat.

[0020] FIG. **3** is a perspective view of a ski tow bar on a boat with a pair of containment assemblies attached thereto, one in an expanded state and one in a collapsed state.

[0021] FIG. **4** is a perspective view of a pair of containment assemblies, one in an expanded state and one in a collapsed state.

[0022] FIG. **5** is a perspective view of a containment assembly in an expanded state with the two lateral pockets empty.

[0023] FIG. **6** is a perspective view of a ski tow bar and containment assembly filled with boating accessories and with the containment assembly attached to upright support portions of the ski tow bar.

[0024] FIG. **7** is a perspective view of a containment assembly in a collapsed state attached to a horizontal portion of a ski tow bar.

[0025] FIG. **8** is a perspective view of a containment assembly in an expanded state and with pockets filled with boating accessories and attached to a ski tow bar.

[0026] FIG. **9** is a perspective view of the back side of a containment assembly in an expanded state illustrating attachment options to a ski tow bar.

[0027] FIG. **10** is a perspective view of a containment assembly with collapsed pockets seated on the deck at a railing of a pontoon boat.

[0028] FIG. **11** is a perspective view of a containment assembly filled with boating accessories and being carried by a shoulder strap on the containment assembly.

DETAILED DESCRIPTION

[0029] Referring to FIG. **1**, a pontoon boat **10** has a starboard pontoon **12** and a port pontoon **14** supporting a deck **20**. The deck **20** is mounted on and above the pontoons and extends in a generally horizontal plane. The deck has decking **21** with a metal framework. The metal framework can include a railing configured as fencing **22** thereon and extending generally around or proximate to the periphery **23** of the deck **20** from the bow **24** to the stern **26** and along the starboard side **27**, and port side **28**. An outboard motor **29** is at the stern. A passenger region **32** is generally defined on the deck and within the fencing **22**. Furniture **35** includes seating, operator's counsel **36**, and storage, which can attach to the decking of the deck. A collapsible canopy **38** can be coupled to fencing **22**, such as one or more railings. A ski tow bar **40** can be attached to the deck **20** at the stern **26** directly forward of the outboard motor. The ski tow bar having a tubular generally horizontal portion **42** and tubular upright support portions **44**, and a tow rope attachment fixture **46**. A containment assembly **50** can be located forward of the ski tow bar **40** and is in an elevated orientation above the deck **20**. Although a pontoon boat is depicted, embodiments also include the containment assembly and ski tow bar in a single hull boat, double hull boat, flat hull boat, displacement hull boat, or the like.

[0030] FIG. **2** is a plan view depicting a slightly differently configured pontoon boat **15** that has a ski tow bar **40'** with generally horizontal portion **42'** and a ski tow rope attachment fixture **46'**. The pontoon boat **15** has a deck **20'**, fencing **22'**, outboard motor **29'**, operator's counsel **36'**, and pontoon boat furniture **35'**. A fully expanded containment assembly **50** with boating accessories **56** therein is attached to the ski tow bar **40'**.

[0031] Referring to FIGS. **3**, **4**, and **5**, different states of a container assembly are illustrated. The containment assembly **50** generally includes a containment portion **51** and an attachment portion **52**. The attachment portion **52** can include at least one adjustable strap to removably secure the containment portion **51** to the framework of the boat. FIG. **3** depicts a ski tow bar **40** with a containment assembly **50**, more specifically the containment portion **51**, in an expanded state and a containment assembly **50'** in a collapsed state. FIG. **4** depicts the expanded containment assembly

50 self-standing with a middle pocket **53**, and two lateral pockets **54**, **55** defining pocket cavities **57**, **58**, and **59** with no contents visible. The lateral side pockets can be positioned external of the containment portion, as the middle pocket **53**. The containment assembly of FIG. **4** in the collapsed state has boating accessories **56** depicted as towels. FIG. **5** has the containment portion **51** in the expanded state with a plurality of internal straps **61**, **62** holding boat fenders **64**, **65** in place in the middle pocket **53**. The containment assembly **50** can be formed of flexible sheet material such as fabric and can be conventionally sewn together to define one or more pockets, such as the three pockets **53**, **54**, and **55**. Each of the pockets **53**, **54**, and **55** can include a plurality of walls that define a cavity extending within the pocket. The cavity can include an open top portion (such as with an upward facing opening), a closed back side, a closed front side, a pair of closed lateral sides, and a closed bottom side. For example, the containment assembly includes a closed first side, a closed second side opposite to the closed first side, a pair of closed lateral sides connecting the first side with the second side; and a closed at least partially closed third side connecting the first side, the second side and the pair of closed lateral sides. In some instances, the closed first side, closed second side and pair of lateral sides defines an open side opposed to the at least partially closed third side. Optionally, the open top portion can be closed with one or more flaps. The one or more flaps can cover, or partially cover, at least one of the one or more open top portions. In other embodiments there can be more or less than the three pockets. As shown in FIG. **5**, each pocket **53**, **54**, and **55** comprises a plurality of soft sided exterior wall portions **72** and one or two interior wall portions **75**, **76**. Each of the interior wall portions **75**, **76**, as illustrated, are part of two pockets, a lateral pocket **54**, **55** and the middle pocket **53**. The fabric wall portions can be reinforced by stiffening material such as cardboard, polymer sheets or other semi rigid materials which can be, for example, sandwiched between two fabric panel portions. The containment assembly of FIG. **4** is depicted as having more rigid lateral pocket exterior wall portions **80** and a more rigid middle pocket exterior wall portions **82**. The containment assembly **50** generally has a front side **85**, a back side **86**, two lateral sides **87**, **88** and a bottom side **89**.

[0032] FIGS. **6** and **7** also depict the containment assembly **50** attached to the framework **90** of a ski tow bar **92** by way of the attachment portion **52**. The ski tow bar is depicted with a substantially horizontal tubular portion **93** and upright tubular support portions **94**, **95** and lower crossing portions **96**. FIG. **8** illustrates the containment assembly **50** on a differently configured ski tow bar **99**. FIG. **9** illustrates the back side **86** of the containment assembly attached to another ski tow bar **101**.

[0033] In an example, the soft sided containment assembly **50** has a detached state and an attached state. In the detached state, the soft sided containment assembly can be decoupled from the metal framework and in the attached state, the soft sided containment assembly can be coupled to the metal framework. In the attached state the soft sided containment assembly can have a fully expanded state with a volumetric capacity larger than a collapsed state.

[0034] Referring to FIGS. **4-9**, the containment assembly **50** has one or more retention straps **104**, **105** extending from the middle pocket **53**, about the respective lateral pockets, for securing the lateral pockets in the folded state of FIGS. **4** and **7**. The one or more retention straps **104**, **105** can be coupled with the flexible material proximate to the one or more pockets. The retention straps **104**, **105** can provide some support for the lateral pockets **54**, **55** when the lateral pockets **54**, **55** are extended, as shown in FIG. **5**. When the lateral pockets are folded up, for example in an accordion fashion, the straps can be tightened for retention of the lateral pockets **54**, **55** in the collapsed state as shown in FIGS. **4** and **7**. The pockets in the collapsed state can be an inch or two in collapsed thickness. More specifically, the thickness of the pockets in the collapsed state can be in a range of about ½ inch to about 3 inches. As best shown in FIG. **9**, the containment assembly **50** also has ski tow bar attachment structure configured as straps **110**, **111**, **112**, **113**, and **114** connecting at a rearward upper margin **115** and rearward lower margin **117** of the containment assembly **50**. The straps **110**, **111**, **112**, **113**, and **114** can be adjustable length straps. The ski tow bar attachment straps

can loop around the generally horizontal upper tubular portion **116** of the ski tow bar **101** and around the lower crossing member **118**. The at least one strap **110**, **111**, **112**, **113**, and **114** is securable about the generally horizontal tubular ski tow bar with a fastener. In an example, the fastener includes at least one of buckles, hook, and loop material, and tieable string or rope. For instance, the at least one strap includes one or more adjustable length upper straps having end portions. The fastener can be positioned at each end portion of at least one of the one or more adjustable length upper straps. The straps have manually operable latches **122** allowing the straps to be opened to be looped around the respective portions of the ski tow bar **99** and then closed to capture same. These vertically extending straps can also be utilized as a backpack type double shoulder straps from transporting the containment assembly on and off the boat on a user's back. As best shown in FIGS. **4** and **11**, a single shoulder strap **125** can attach at the upper edge portion **126** of the three pockets of the containment assembly **50** to allow a fully loaded containment assembly to be readily carried generally at a waist level. The boating accessories suitable for carrying in the containment assembly **50** include towels **63**, boat fenders **64**, **65** and life jackets **129**. These accessories are relatively bulky but with low weight density. The containment assembly allows a variable volumetrically sized bag system with a variable number of pockets for different loads of boating accessories. Referring to FIG. **11**, the shoulder strap **125** permit a person **131** to easily carry towels **63**, boat fenders **64**, **65**, and life jackets **129** and still have his hands free. Referring to FIG. **5**, multiple attachment points defined by a plurality of eyelets or polymer loops **132** attached along an upper margin of the pockets can allow attaching the shoulder strap at different positions. In embodiments, the containment assembly **50** fully loaded with life jackets, towels, and boat fenders, for example, can weigh less than about **30** pounds and is thus readily carried with the shoulder strap **125**.

[0035] Referring to FIGS. **4** and **5**, in embodiments, the containment assembly **50** has a front panel **134** with three panel portions **136**, **137**, and **138** associated with the three pockets **53**, **54**, **55**, a rear panel **142** with three panel portions **144**, **145**, and **146** also associated with the three pockets. The length L of the containment assembly **50**, can be defined as the length of the front panel **134**, see double arrow. The length L can be measured where the panel has rigid structure. Where the panels are flexible fabric material, the length L can be measured when the panel is pulled taught. In embodiments, the length of the containment assembly can be a length of the rear panel **142**. In embodiments, the length L, can be in the range of about 20 inches to about 40 inches. In embodiments, four bridging panels **150**, **152**, **153**, and **154** extend between the front panel **134** and rear panel defining the three pockets. In embodiments, the pocket depth PD of the containment assembly, defined as the length of longest bridging panel or panels, can be in the range of about 5 inches to 14 inches. In embodiments, the pocket depth PD of the containment assembly can be in the range of about 8 inches to 12 inches. In embodiments, the pocket depth PD of the containment assembly can be in the range of about 6 inches to 20 inches. In embodiments, the containment assembly **50** can have supplemental compartments **157**, **158** with closures such as zippers **160**. A depth D of the containment assembly **50** including such supplemental pockets in a full expanded state can be in the range of 9 inches to about 18 inches. In embodiments, the depth D can be in the range of about 12 to 24 inches. In embodiments, the height H of the containment assembly, equating to the depth of the pockets, can be in the range of about 10 inches to about 28 inches. In embodiments, the height H of the containment assembly, equating to the depth of the pockets, can be in the range of about 13 inches to about 18 inches.

[0036] Referring to FIGS. **2** and **10**, the containment assembly **50** can also be seated on the boat deck and the straps can be attached to boat structure such as fencing **22**. Multiple locations **170**, indicated in dashed lines on FIG. **2**, are suitable for receiving the containment portion **51**, in an expanded, partially collapsed, or collapsed state allowing attachment to the fencing and if desired, elevation of the containment assembly **50**.

[0037] In embodiments, the total volume of the upwardly opening pockets of the containment

assembly is in the range of about 1.5 to about 3 cubic feet. Volume defined as the internal volumetric capacity of the pocket cavity up to the upper margin of the pocket, which can define the pocket opening, in the expanded pocket. In a rectilinear pocket, the volume is the internal height×the internal depth×the internal width of the expanded pocket. In embodiments, the total volume of the upwardly opening pockets of the containment assembly is in the range of about 1.8 to about 4 cubic feet. In embodiments, the volume of a single pocket or the cumulative volume of a plurality of pockets is greater than about 2 cubic feet. In embodiments, an upwardly opening single pocket of a containment portion 51 can have a volume in the range of about 0.4 to about 4.0 cubic feet. In embodiments, a containment portion 51 with two pockets, each pocket can have a volume in the range of about 0.4 to about 2 cubic feet. When a particular pocket is collapsed, the volume of that pocket becomes zero. The volumetric displacement of the collapsed pocket material when in the collapsed state, can be less than 15% of the expanded volume capacity. In embodiments, rather than having a plurality of pockets with internal wall portions dividing the pockets, the containment assembly can have one large pocket with pocket dividers configured as straps for example. In such a case the volume of the single pocket can be greater than about 1.5 cubic feet. The internal straps can secure the contents and can be adjustable in length to pull exterior wall portions together reducing the volume of the pocket providing enhanced securement integrity of the contents.

[0038] In embodiments, two or more containment assemblies can be connected utilizing the straps with fasteners for interconnecting the containment assemblies to increase the capacity.

[0039] In some aspects, the techniques described herein relate to a boat storage system including: a ski tow bar including a ski tow bar framework, the framework having a generally horizontal upper tubular portion supported by a plurality of upright tubular support portions; a containment assembly including a containment portion and an attachment portion integrated with the containment portion, the containment portion having one or more pockets formed of a flexible material, the attachment portion including at least one adjustable strap extending about and removably secured to ski tow bar framework.

[0040] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment assembly includes a plurality of material panels sewn together to define the containment assembly.

[0041] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment assembly has an open top, a closed back side, a closed front side, a pair of closed lateral sides, and a closed bottom side.

[0042] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment assembly has a reduced volume collapsed state and a full volume expanded state, wherein the containment assembly can be retained in the reduced volume state by one or more retention straps securing one or more of the plurality of pockets in a folded state, wherein in the full volume state, the containment assembly has a volumetric capacity of at least one cubic foot.

[0043] In some aspects, the techniques described herein relate to a boat storage system, wherein the at least one strap is securable about the generally horizontal tubular ski tow bar by at least one of buckles, hook and loop material, and tieable string or rope.

[0044] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment assembly has one or more flaps for covering the one or more pockets.

[0045] In some aspects, the techniques described herein relate to a boat storage system, further including a boat to which the ski tow bar is mounted, and the ski tow bar supports the containment assembly above and spaced from a deck of the boat and wherein the containment assembly projects forwardly from the ski tow bar.

[0046] In some aspects, the techniques described herein relate to a boat storage system, further including a plurality of boating accessories received in the plurality of pockets, the boating accessories selected from the set of boat fenders, towels, life jackets.

[0047] In some aspects, the techniques described herein relate to a boat storage system, wherein the

containment assembly further includes a shoulder strap of at least three feet in length with each of two ends attached or attachable to a top edge portion of the containment portion.

[0048] In some aspects, the techniques described herein relate to a boat storage system, wherein the flexible material includes a fabric.

[0049] In some aspects, the techniques described herein relate to a boat storage system, wherein there are a plurality of adjustable straps, each extending about and removably secured to ski tow bar framework each of the plurality of adjustable straps are adjustable in length and are buckleable.

[0050] In some aspects, the techniques described herein relate to a boat storage system including: a boat with a metal framework having a generally horizontal upper framework portion supported by a plurality of upright framework support portions; a soft sided containment assembly mounted to the generally horizontal upper tubular member, the containment assembly including a containment portion formed of a flexible material, the containment portion having an upward opening for receiving boating accessories in the containment portion, the soft sided containment assembly further including a flexible attachment portion attached to an upper edge portion of the containment portion, the flexible attachment portion is secured to and extends over the generally horizontal upper framework portion, the flexible attachment portion securable to the framework and removable from the framework manually without tools.

[0051] In some aspects, the techniques described herein relate to a boat storage system, wherein the flexible attachment portion includes a plurality of adjustable length straps with buckles, each of the plurality of adjustable length straps extending to the framework.

[0052] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment portion of the soft sided containment assembly has a fully expanded state with a volumetric capacity of at least 2 cubic feet.

[0053] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment portion defines a plurality of pockets with top openings, the plurality of pockets formed of the flexible material.

[0054] In some aspects, the techniques described herein relate to a boat storage system, wherein the plurality of pockets with top openings are five sided pockets formed of the flexible material, and wherein one or more of the five sided pockets is individually configurable between a collapsed state and a fully expanded state and is retainable in the collapsed state.

[0055] In some aspects, the techniques described herein relate to a boat storage system, wherein the one of more of the plurality of five sided pockets are individually retainable in the collapsed state by way of a strap with a buckle.

[0056] In some aspects, the techniques described herein relate to a boat storage system, wherein the framework is a ski tow bar framework.

[0057] In some aspects, the techniques described herein relate to a boat storage system, wherein the attachment portion of the containment assembly includes a plurality of adjustable length upper buckleable straps at a rearward upper edge portion of the containment portion and another plurality of adjustable length lateral buckleable straps at two lateral portions of the containment portion, and wherein the plurality of adjustable length upper buckleable straps extend about the generally horizontal upper framework portion and the plurality of adjustable length lateral buckleable straps extend about the plurality of upright framework support portions.

[0058] In some aspects, the techniques described herein relate to a boat storage system for stowing boating accessories, the boat storage system including: a boat with a metal framework extending upwardly above a deck of the boat, the framework having an upper framework portion elevated above the deck, the upper framework portion supported by a plurality of upright support framework portions; a soft sided containment assembly including a containment portion with a plurality of pockets, and an attachment portion attached to the containment portion, the attachment portion including one or more adjustable length buckleable upper straps and a plurality of adjustable length buckleable lateral straps, wherein the one or more adjustable length buckleable upper straps are

secured to the upper framework portion and wherein one or more of the plurality of adjustable length buckleable lateral straps are secured to one or more of the plurality of upright support framework portions.

[0059] In some aspects, the techniques described herein relate to a boat storage system, wherein the boat includes pontoons supporting a pontoon deck, and wherein the metal framework extends from a pontoon deck.

[0060] In some aspects, the techniques described herein relate to a boat storage system, wherein the framework is fencing extending about a passenger region of the pontoon deck.

[0061] In some aspects, the techniques described herein relate to a boat storage system, wherein the framework is a ski tow bar framework, and wherein the upper framework portion has a ski rope attachment fixture secured thereto.

[0062] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment portion is formed of a flexible fabric material, and the containment portion includes a plurality of fabric panels sewn together to define one or more upwardly opening pockets.

[0063] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment portion defines a plurality of pockets, each with a respective pocket cavity and a top opening, the plurality of pockets formed of a flexible fabric material.

[0064] In some aspects, the techniques described herein relate to a boat storage system, wherein the plurality of pockets with top openings are five sided pockets formed of the fabric material, and wherein one or more of the five sided pockets is individually configurable between a collapsed state and a fully expanded state and is retainable in the collapsed state.

[0065] In some aspects, the techniques described herein relate to a boat storage system, wherein the one or more of the plurality of pockets are individually retainable in the collapsed state by way of a respective strap with a buckle.

[0066] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment portion of the soft sided containment assembly has a fully expanded state with a volumetric capacity of at least 2 cubic feet.

[0067] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment assembly is elevated above the deck of the boat due to an attachment position of the attachment portion on the framework.

[0068] In some aspects, the techniques described herein relate to a boat storage system, wherein the containment portion of the containment assembly is loaded with boat accessories including one or more boat fenders, one or more life jackets, and/or one or more towels, the one or more boat fenders, one or more life jackets, and/or one or more towels visually exposed with respect to a passenger on the deck of the boat and manually removable from the containment portion by the passenger on the deck of the boat.

[0069] In some aspects, the techniques described herein relate to a boat storage system, wherein the one or more boat fenders, the one or more life jackets, and/or the one or more towels can be removed from the containment portion without opening a closure on the containment portion.

[0070] In some aspects, the techniques described herein relate to a boat storage system with boating accessories stowed therein, the boat storage system including: a boat with a metal framework extending upwardly above a deck of the boat, the framework having an upper framework portion elevated above the deck, the upper framework portion supported by a plurality of upright support framework portions; a soft sided containment assembly including a containment portion and an attachment portion, the containment portion holding the boating accessories, the attachment portion securing the containment assembly to the metal framework.

[0071] In some aspects, the techniques described herein relate to a boat storage system, wherein the attachment portion includes one or more adjustable length buckleable straps.

[0072] In some aspects, the techniques described herein relate to a boat storage system, wherein the attachment portion includes a plurality of adjustable length buckleable upper straps and a plurality

of adjustable length buckleable lateral straps, wherein the one or more adjustable length buckleable upper straps are secured to the upper framework portion and wherein one or more of the plurality of adjustable length buckleable lateral straps are secured to one or more of the plurality of upright support framework portions.

[0073] In some aspects, the techniques described herein relate to a boat storage system, wherein the volumetric capacity of the containment portion is at least 2 cubic feet.

[0074] In some aspects, the techniques described herein relate to a boat storage system, wherein the container portion has a plurality of pockets and at least one of the plurality of pockets is convertible from a fully expanded state to a collapsed state and wherein the at least one of the plurality of pockets can be retained in the collapsed state by way of a strap.

[0075] In some aspects, the techniques described herein relate to a boat storage system, wherein the boating accessories are visually exposed in the container portion and are removable therefrom without opening any closures of the containment portion.

[0076] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, the boat storage system including: a containment assembly formed of flexible sewn fabric and having a row of three pockets including two lateral pockets and a middle pocket, each of the pockets having an upper opening, the containment assembly having an expanded state with each of the three pockets expanded to a full capacity for receiving the boating accessories, and a collapsed state with at least the two lateral pockets in a folded-in state, the two lateral pockets retainable in the folded-in state, the row pockets of the containment assembly, when in the expanded state, having a width of at least 22 inches, a height of at least 13 inches, and a depth of at least 8 inches, the containment assembly further including a plurality of upper attachment straps proximate an upper margin of a back side of the sewn containment assembly for attachment of the containment assembly to a horizontal portion of a ski tow bar.

[0077] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly further includes a plurality of lower attachment straps with attachment points in horizontal alignment positioned on the backside of the sewn containment assembly closer to a lower margin of the backside than to the upper margin.

[0078] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the two lateral pockets can be retained in the folded-in state with a pair of retention straps attached to the middle pocket.

[0079] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, including two boating accessories, the boating accessories including a pair of boat fenders at least 20 inches long and having a middle diameter of at least about 6 inches.

[0080] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, including packaging of the pair of boat bumpers, for the containment assembly, and for instructions for use in the packaging.

[0081] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, further including a ski tow bar with a tubular upper portion and a pair of tubular support portions connecting to the tubular upper portions and extending to respective mounting flanges, the tubular upper portion positioned at least about 24 inches from the respective mounting flanges, and wherein the containment assembly is attached to or attachable to the ski tow bar with the plurality of attachment straps extending around the tubular upper portion.

[0082] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the ski tow bar is mounted to a boat at or proximate to the stern of the boat and wherein the containment assembly is mounted to the ski tow bar, and containment assembly is raised off of a deck of the boat.

[0083] In some aspects, the techniques described herein relate to a boat storage system, wherein the ski tow bar has a lower crossing portion extending between the two tubular support portions and wherein the containment assembly is attached to the lower crossing portion lower attachment

straps.

[0084] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the ski tow bar and the containment assembly are in a retail assemblage including packaging, the retail assemblage including instructions for use.

[0085] In some aspects, the techniques described herein relate to the boat storage system for boating accessories of any of the above claims, wherein each of the pockets of the containment assembly do not have covers.

[0086] In some aspects, the techniques described herein relate to the boat storage system for boating accessories of any of the above claims, further including one or more closeable compartments on a front side of the middle pocket, each closeable compartment with an internal volume of at least about 0.20 cu. ft.

[0087] In some aspects, the techniques described herein relate to the boat storage system for boating accessories of any of the above claims, wherein the containment assembly further has a strap attached or attachable thereto for manually carrying the containment assembly with boating accessories therein off and on a boat.

[0088] In some aspects, the techniques described herein relate to the boat storage system for boating accessories of any of the above claims further including a plurality of boating accessories in the containment assembly, the boating accessories selected from the set of boat fenders, towels, life jackets.

[0089] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly is mounted to the ski tow bar on the boat and a plurality of boating accessories in the containment assembly are projecting upwardly out of one or more pockets of the containment assembly.

[0090] In some aspects, the techniques described herein relate to the boat storage system for boating accessories of any of the above claims wherein the containment assembly has drainage openings at a bottom of the containment assembly.

[0091] In some aspects, the techniques described herein relate to the boat storage system for boating accessories of any of the above claims, wherein one or more of the plurality of upper attachment straps and plurality of lower attachment straps have elastic portions.

[0092] In some aspects, the techniques described herein relate to the boat storage system for boating accessories of any of the above claims, wherein the weight of the containment assembly without any accessories is less than 3 lbs.

[0093] In some aspects, the techniques described herein relate to the boat storage system for boating accessories of any of the above claims, wherein one or more of the three pockets have closable flap to cover the respective open tops of the respective pockets.

[0094] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly is mounted to a railing on a boat.

[0095] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, the boat storage system including: a containment assembly attached to a ski tow bar on a boat; the containment assembly having a row of a plurality of compartments with open tops for receiving boating accessories, a plurality of upper attachment straps extending from a back side of the containment assembly to the ski tow bar, a plurality of lower attachment straps extending from the back side of the containment assembly below the upper attachment straps to the ski tow bar, the containment portion having a volumetric capacity defined by the plurality of compartments of at least about 2 cubic feet, the containment assembly elevated at least about 5 inches above boat structure to which the ski tow bar is attached.

[0096] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly has a plurality of flexible fabric wall portions defining the compartments.

[0097] In some aspects, the techniques described herein relate to a boat storage system for boating

accessories, wherein the plurality of compartments include three compartments in a row.

[0098] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein one or more of the plurality of compartments have rigid side walls.

[0099] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly has one or more zippered pockets on an exterior surface of the plurality of compartments.

[0100] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly contains a plurality of boating accessories projecting upwardly from the plurality of compartments.

[0101] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly is formed of flexible sewn fabric.

[0102] In some aspects, the techniques described herein relate to a boat storage system for boating accessories wherein the containment assembly extends forwardly from the ski tow bar toward the bow of the boat.

[0103] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly is collapsible to a reduced size.

[0104] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly has a shoulder strap for carrying the containment assembly with accessories on and off the boat.

[0105] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein at least one of the compartments has a divider strap extending between a forward wall portion and a rearward wall portion of the respective pocket.

[0106] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the containment assembly is attachable to and removable from the ski tow bar without tools.

[0107] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, the boat storage system including: a boat with a tow bar and an expandable-collapsible boating accessory containment assembly removably attached to the tow bar; wherein the tow bar having an upper horizontal tubular portion extending in a port-starboard direction, a pair of downward tubular portions extending to and being attached to the boat, a ski rope attachment fixture attached to or attachable to a center location on the upper horizontal tubular portion; wherein the expandable-collapsible containment assembly has a fully expanded state and one or more collapsed states, the containment assembly including: a plurality of exterior wall portions and one or more interior divider wall portions, each of the exterior wall portions attached to and extending from a bottom side wall portion, and wherein each of the plurality of exterior wall portions and interior divider wall portions are formed of a flexible fabric, the plurality of exterior wall portions and interior divider wall portions defining a plurality of five-sided pockets, each four sided pockets having a respective upward pocket opening, wherein when the containment assembly is in the fully expanded states the volume of each of the plurality of the five sided pockets having a volume of at least about 0.4 cubic foot; wherein one of the exterior wall portions defining an attachment side wall portion of the containment assembly, the containment assembly further including a plurality of upper attachment straps extending from an upper region of the attachment side wall portion, the plurality of upper attachment straps removably attached to the upper tubular horizontal portion; the containment assembly further including a plurality of lower attachment straps extending from one or more of the plurality of exterior wall portions, the plurality of lower attachment straps each attached to the tow bar below the upper horizontal tubular portion.

[0108] In some aspects, the techniques described herein relate to a boat storage system for boating accessories, wherein the plurality of five sided pockets include a central pocket intermediate two lateral pockets, and wherein each of the two lateral pockets are collapsible to a flattened configuration, and wherein the containment assembly further includes a pair of retention elements

for securing each of the two lateral pockets in the flattened configuration.

[0109] While the disclosure has been described in connection with what is presently considered to be the most practical and preferred embodiments, it is to be understood that the disclosure is not to be limited to the disclosed embodiments, but on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as permitted under the law. Furthermore, while the use of the word preferable, preferably, or preferred in the description above indicates that feature so described can be more desirable, it nonetheless may not be necessary and any embodiment lacking the same may be contemplated as within the scope of the disclosure, that scope being defined by the claims that follow. In reading the claims it is intended that when words such as “a,” “an,” “at least one” and “at least a portion” are used, there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. Further, when the language “at least a portion” and/or “a portion” is used the item may include a portion and/or the entire item unless specifically stated to the contrary.

Claims

1. A boat storage system comprising: a containment assembly including: a containment portion including one or more pockets formed of a flexible material; and an attachment portion integrated with the containment portion, wherein the attachment portion includes at least one adjustable strap configured to extend about and removably secure the containment portion to a framework of a boat.
2. The boat storage system of claim 1, wherein the containment assembly includes a plurality of material panels sewn together to define the containment assembly.
3. The boat storage system of claim 1, wherein the containment assembly includes: a closed first side; a closed second side opposite to the closed first side; a pair of closed lateral sides connecting the first side with the second side; and a closed at least partially closed third side connecting the first side, the second side and the pair of closed lateral sides; wherein the closed first side, closed second side and pair of lateral sides defines an open side opposed to the at least partially closed third side.
4. The boat storage system of claim 1, including one or more retention straps coupled with the flexible material proximate to the one or more pockets; wherein the containment assembly includes a full volume expanded state having a larger volumetric capacity than a reduced volume collapsed state; wherein the containment assembly is configured to be retained in the reduced volume collapsed state by the one or more retention straps.
5. The boat storage system of claim 1, wherein the at least one adjustable strap configured to be secured about a horizontal tubular ski tow bar.
6. The boat storage system of claim 1, wherein the containment assembly has one or more flaps configured to cover at least one of the one or more pockets.
7. The boat storage system of claim 1, wherein the containment assembly includes a shoulder strap coupled to a top edge portion of the containment portion.
8. A boat storage system comprising: a boat with a metal framework having a horizontal upper framework portion supported by a plurality of upright framework support portions; a soft sided containment assembly mounted to the horizontal upper framework portion, the soft sided containment assembly including: a containment portion formed of a flexible material, the containment portion having an upward facing opening; a flexible attachment portion attached to an upper edge portion of the containment portion; wherein the flexible attachment portion is removably securable to the horizontal upper framework portion.
9. The boat storage system of claim 8, wherein the flexible attachment portion includes a plurality of adjustable length straps; wherein each of the plurality of adjustable length straps is coupled with

a portion of the soft sided containment assembly facing the metal framework.

10. The boat storage system of claim 8, wherein the soft sided containment assembly has a detached state and an attached state; wherein in the detached state, the soft sided containment assembly is decoupled from the metal framework and in the attached state, the soft sided containment assembly is coupled to the metal framework; wherein in the attached state the soft sided containment assembly has a fully expanded state with a volumetric capacity larger than a collapsed state.

11. The boat storage system of claim 8, wherein the soft sided containment assembly defines a plurality of pockets with top openings, the plurality of pockets formed of a flexible material.

12. The boat storage system of claim 8, including one or more pockets formed within the soft sided containment assembly; wherein each pocket of the one or more pockets is individually configurable between a collapsed state and an expanded state.

13. A boat storage system for stowing boating accessories, the boat storage system comprising: a boat with a metal framework extending upwardly above a deck of the boat, the metal framework including: a plurality of upright support framework portions extending from the deck; and an upper framework portion supported by at least two of the plurality of upright support framework portions, the upper framework portion elevated above the deck; and a soft sided containment assembly comprising a containment portion, the containment portion defined by four walls extending around a cavity, the soft sided containment assembly includes: a plurality of lateral pockets defined by the containment portion and positioned within the cavity; at least one supplemental compartment positioned on an exterior facing surface of at least one wall of the containment portion; and an attachment portion coupled to an exterior surface of at least one wall different from the at least one wall with the at least one supplemental compartment; wherein the attachment portion includes one or more adjustable length upper straps and one or more adjustable length lateral straps; wherein the one or more adjustable length upper straps are configured to be secured to the upper framework portion and wherein another of one or more of the one or more adjustable length lateral straps is configured to be secured to one or more of the plurality of upright support framework portions.

14. The boat storage system of claim 13, wherein the metal framework includes a ski tow bar framework.

15. The boat storage system of claim 13, wherein the soft sided containment assembly configured to be coupled in an elevated orientation above the deck.

16. The boat storage system of claim 13, wherein the containment portion is formed of a flexible material; wherein the containment portion includes one or more laterally extending panels, extending within the cavity.

17. The boat storage system of claim 13, wherein the soft sided containment assembly includes one or more side pockets positioned external of the containment portion; wherein the one or more lateral pockets are configured to have an expanded configuration and a collapsed configuration; wherein in the collapsed configuration, the one or more lateral pockets have a reduced volume than in the expanded configuration.

18. The boat storage system of claim 13, including one or more drainage openings formed within a bottom surface of containment portion.

19. The boat storage system of claim 13, wherein the one or more adjustable length upper straps includes a fastener positioned at each end portion of at least one of the one or more adjustable length upper straps.

20. The boat storage system of claim 13, wherein the attachment portion includes one or more lower straps, the lower straps including a fastener positioned at each end portion of at least one of the one or more lower straps.
